

Installing Antumbra on the OnePlus 7T Pro

Step by step, from a phone running OxygenOS to Antumbra's Welcome screen, using a Windows, macOS or Linux computer.

About 2 hours

including downloads and two phone resets

Erases the phone

back up first: everything on it is deleted

HD1913 (Europe)

the model the underlying port was tested on

Alpha software

not yet booted on a phone by its authors

READ THIS BEFORE YOU START

Antumbra is **alpha software that its authors have not yet started on a real phone**. It is built on a Linux port for this phone that one developer has tested on hardware, and Antumbra itself has been tested only in a virtual machine. Expect problems: the phone may not start Antumbra at all, Wi-Fi, sound or graphics may not work, and you may have to reinstall Android.

Installing **erases everything on the phone**, twice: once when you unlock the bootloader, and again when Antumbra is written. Use a phone you can afford to be without, and follow the steps in order. If the phone is ever recognised by your computer as QDLoader 9008 or QUSB_BULK, stop and read section 12.

1 What you need

2 Check your phone's model

3 Set up your computer

4 Back up, F.22 in both slots

5 Unlock the bootloader

6 Download and check Antumbra

7 Prepare the install

8 Install Antumbra

9 First start

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11 Going back to Android

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A Quick reference

1 What you need

STEPS 2–3

Check the model, install fastboot on the computer

STEP 4

Back up, put OxygenOS F.22 in both slots

STEP 5

Unlock the bootloader (erases the phone)

STEPS 6–7

Download, check, find the free slot

STEPS 8–9

Write Antumbra, start it

Item	Details
The phone	A OnePlus 7T Pro, ideally model HD1913 (section 2), charged to well over half. Not the OnePlus 7T, 7 Pro or the T-Mobile 7T Pro 5G McLaren.
A USB cable	A USB-C cable that carries data, not only charge (the one that came with the phone is best). If your computer has a USB-A port, a USB-A to USB-C cable in a USB 2.0 port is the most reliable.
A computer	Windows 10 or 11, macOS, or Linux, with about 5 GB of free space and an internet connection. You will type a few commands into a terminal window; this guide gives every one of them.
A GitHub account	The Antumbra files are on the draft release of your GitHub repository, which only you can see while you are signed in (section 6).
Time	About two hours. Do not start shortly before you need the phone.

Words this guide uses

Bootloader	The small program that starts the phone. In its fastboot mode your computer can write to the phone's storage.
fastboot	The command-line tool on your computer that talks to the bootloader, part of Google's free platform-tools . fastbootd is a second fastboot mode, provided by the phone's recovery, which can write the large Antumbra file.
Slot A / slot B	The phone keeps two sets of its start-up files, called slots. Android runs from one of them; Antumbra goes into the other, so Android's own slot stays untouched and you can go back.
Firmware	Programs that run inside the phone's chips (Wi-Fi, modem, sound, graphics). They belong to Qualcomm and OnePlus and are already on your phone. Antumbra's files contain none of them: at every start Antumbra reads them, read-only, from the phone's own storage.

WHAT ANTUMBRA CAN AND CANNOT DO ON THIS PHONE

Expected to work (as on the Linux port it is built on): the screen at 60 Hz, touch, Wi-Fi, the speakers, USB-C, the cameras (basic photos), and Antumbra's own features: Tor for all connections, the Welcome screen, optional encrypted Persistent Storage, Tor Browser, and optional Android apps.

Not working: phone calls, SMS and mobile data (the modem only runs for Wi-Fi), the fingerprint reader, automatic screen rotation, the earpiece and headsets, 90 Hz, fast charging. Charging from a computer's USB port is slow; use a wall charger.

2 Check your phone's model

Look at the **model number on the box or on the label** (on the back of the phone or under the SIM tray area). You can also open **Settings** → **About phone** (on OxygenOS 12: **Settings** → **About device**), but if the phone has ever run LineageOS, Settings can show **HD1911** whatever the real model is.

Model	Where it was sold	Can you use this guide?
HD1913	Europe	Yes. The Linux port Antumbra is built on was tested on this model.
HD1911	India	Same software line, but the full Linux port has never been started on one. You can try at your own risk.
HD1917	North America (unlocked)	As for HD1911: untested.
HD1910	China	As for HD1911: untested.
HD1925	T-Mobile (7T Pro 5G McLaren)	No. A different phone, and its bootloader needs an unlock token from OnePlus.

WHY THIS MATTERS

The commands in this guide write to fixed places on the phone's storage. On a different phone they could damage it. The install command checks the size of every place it writes to and stops if one does not match, but that is a safety net, not a reason to skip this check.

3 Set up your computer

You need Google's **SDK Platform-Tools**, which contain adb and fastboot. Download them only from Google's page: developer.android.com/tools/releases/platform-tools.

Windows 10 or 11

1. Download **SDK Platform-Tools for Windows** and extract the zip to `C:\platform-tools` (so that `C:\platform-tools\fastboot.exe` exists).
2. Download Google's **USB driver** (developer.android.com/studio/run/win-usb, the zip file) and extract it to `C:\usb_driver`. You will need it in section 5.
3. Open the `C:\platform-tools` folder, right-click an empty area and choose **Open in Terminal** (Windows 11) or hold **Shift** and right-click, then **Open PowerShell window here** (Windows 10).
4. Type `.\fastboot.exe --version` and press **Enter**. A version number means it works.

Important on Windows: in PowerShell every command from the platform-tools folder starts with `.\`. Where this guide writes `fastboot devices`, you type `.\fastboot.exe devices`, and `adb` becomes `.\adb.exe`.

macOS

1. Download **SDK Platform-Tools for Mac**; the zip unpacks to a `platform-tools` folder in **Downloads**.
2. Open **Terminal** (in **Applications** → **Utilities**) and type `cd ~/Downloads/platform-tools`.
3. Type `./fastboot --version`. If macOS refuses to open it, allow it in **System Settings** → **Privacy & Security** and try again. (With Homebrew, `brew install --cask android-platform-tools` works too; then type the commands without `./`.)

Where this guide writes `fastboot`, type `./fastboot` (and `./adb` for `adb`).

Linux

1. Download **SDK Platform-Tools for Linux** and unzip it into your home folder, or install your distribution's package (for example `android-tools` or `fastboot`; it must be version 31 or newer).
2. Open a terminal in the `platform-tools` folder and type `./fastboot --version`.
3. If later `fastboot devices` prints **no permissions**, install the `android-udev-rules` package, or run `fastboot` with `sudo`.

Where this guide writes `fastboot`, type `./fastboot`. Linux users who have the Antumbra repository can instead run its guided script, which performs sections 7 and 8 with checks: `build/flash.sh --release FOLDER`.

KEEP EVERYTHING IN ONE FOLDER

In section 6 you will put the Antumbra files into this same `platform-tools` folder. Then every command in this guide works as written, without typing long paths.

4 Back up, then put OxygenOS 12 F.22 in both slots

Antumbra runs on the start-up software and firmware of its own slot, and the Linux port it is built on was tested with OxygenOS 12 build **F.22**, the last one for this phone. A system update only writes the slot Android is **not** using and then switches to it, so after one update the other slot still holds the previous version. Installing F.22 twice puts it in both.

- 1 **Back up everything you want to keep:** photos, messages, two-factor codes, contacts. Section 5 erases the phone completely.
- 2 **Turn on Developer options and USB debugging.** On OxygenOS 12: **Settings** → **About device** → **Version**, tap **Build number** seven times (enter your PIN if asked); on older versions: **Settings** → **About phone**, tap **Build number** seven times. Then open **Developer options** (search Settings for it) and switch on **OEM unlocking** and **USB debugging**.
- 3 **Connect the phone** and type `adb devices`. On the phone, allow USB debugging for this computer (tick **Always allow**). The command lists the phone with the word **device**.
- 4 **Update to F.22:** **Settings** → **System** → **System updates**, installing every update offered. The version (**Settings** → **About device**) then reads something like HD1913_11_F.22.
- 5 **Note the current slot:** type `adb shell getprop ro.boot.slot_suffix`. It prints `_a` or `_b`.
- 6 **Install F.22 a second time**, from the full update package: download it with the free **Oxygen Updater** app (it fetches OnePlus's own files), then use **Local install**, which on OxygenOS 12 is in the menu (top right) of **Settings** → **About device** → **Up to date** once Developer options are on. Let it finish and restart.
- 7 **Check that it worked:** `adb shell getprop ro.boot.slot_suffix` now prints the **other** letter. Both slots then hold F.22.
- 8 **Restart the phone once more** and wait a few minutes with the screen on: OxygenOS 12 finishes an update in the background after the first restart, and the phone refuses to install Antumbra until it has (section 7 checks this).

IF THE SECOND INSTALL IS REFUSED

Nobody has confirmed that OxygenOS accepts installing the version it already runs. If it refuses, or the letter in step 7 did not change, you can continue: Antumbra will then run on the start-up software and firmware of the previous OxygenOS version, a combination nobody has tested. If Antumbra does not start, going back to Android (section 11) works either way.

5 Unlock the bootloader

THIS ERASES THE PHONE

Unlocking deletes everything on the phone. After it, the phone shows a warning that the bootloader is unlocked every time it starts; that is normal and stays. OnePlus says an unlocked phone no longer receives official updates (there are none left for this phone). Never lock the bootloader again while Antumbra is installed.

- 1 **Check** that **OEM unlocking** and **USB debugging** are still on in **Developer options** (section 4), and connect the phone.
- 2 **Start the bootloader:** type `adb reboot bootloader`. (Without a computer: switch the phone off, then hold **Power** + **Volume Up** + **Volume Down** until the bootloader screen appears.)
- 3 **Check the connection:** type `fastboot devices`. It should print one line with a serial number and the word `fastboot`.

WINDOWS: NOTHING PRINTED?

Windows does not know the bootloader's USB device yet. Open **Device Manager** (right-click Start). Find the phone, usually under **Other devices** with a yellow triangle (often called **Android**). Right-click it, **Update driver** → **Browse my computer for drivers** → **Let me pick from a list** → **Have Disk...**, browse to `C:\usb_driver` and pick `android_winusb.inf`, then choose **Android Bootloader Interface** and accept the warning. Run `fastboot devices` again. Restart the computer if it still shows nothing.

- 4 **Unlock:** type `fastboot oem unlock`. The phone asks whether to unlock. Use the **Volume** keys to choose the unlock option and press **Power** to confirm.
- 5 The phone erases itself and restarts into Android's first-time setup. Go through it quickly (no accounts are needed). The bootloader is now unlocked: you can check with `fastboot getvar unlocked` in the bootloader, which should say `unlocked: yes`.

6 Download and check Antumbra

- 1 Sign in to GitHub and open github.com/jaivik-2612/vscode_remote/releases. The draft release "**Antumbra 0.1.0-alpha.2 for the OnePlus 7T Pro**" is visible only to you (and to anyone you give write access). You do not need to publish it.

- 2 Under **Assets**, download these files into your `platform-tools` folder:

File	Size	What it is
<code>antumbra-0.1.0-alpha.2-oneplus-hotdog-boot.img</code>	96 MB	Antumbra's start-up image
<code>antumbra-0.1.0-alpha.2-oneplus-hotdog-userdata.img</code>	about 1.5 GB	the system itself
<code>antumbra-0.1.0-alpha.2-oneplus-hotdog-dtbo.img</code>	24 MB	hardware description
<code>antumbra-0.1.0-alpha.2-oneplus-hotdog-vbmeta-disabled.img</code>	64 KB	boot verification settings
<code>SHA256SUMS</code>	1 KB	the files' checksums
<code>MANIFEST.md</code> , <code>INSTALL.md</code>	small	what is inside; the technical version of this guide

The kernel source is there too; you do not need it to install. Never mix files from different releases.

- 3 **Check that the downloads are complete and unchanged.** In the terminal, in the `platform-tools` folder:

macOS

```
shasum -a 256 -c SHA256SUMS
```

Linux

```
sha256sum -c SHA256SUMS
```

Every line must end in OK.

Windows (PowerShell), one line per file:

```
Get-FileHash .\antumbra-0.1.0-alpha.2-oneplus-hotdog-boot.img
```

Open `SHA256SUMS` in Notepad and compare the long number printed under **Hash** with the one on the file's line (capital or small letters do not matter). Do this for all four `.img/.img` files.

If a file does not match, download it again. Never write a file that does not match.

7 Prepare the install

These checks change nothing on the phone. They find out which slot Android uses, and make sure the phone can receive the large system file.

- 1 **Start the bootloader:** switch the phone off, then hold **Power** + **Volume Up** + **Volume Down** until the bootloader screen appears. Connect it and type `fastboot devices` (one line).
- 2 **Check the unlock:** `fastboot getvar unlocked` must answer `unlocked: yes`.
- 3 **Find Android's slot:** type `fastboot getvar current-slot`. It answers `current-slot: a` or `current-slot: b`. **Write the letter down and keep the note:** it is Android's slot. Antumbra goes into the **other** slot, called X in this guide, now and for every later install:

Android's slot	Antumbra's slot X
a	b
b	a

After the install, `current-slot` shows Antumbra's slot, so it can no longer tell you which one is Android's. That is why you need the note.

- 4 **Start fastbootd** (the recovery's fastboot mode): type `fastboot reboot fastboot`. The phone restarts into a screen titled **fastbootd** or similar. Then type `fastboot getvar is-userspace`: it must answer `is-userspace: yes`.

Windows: if the command waits forever (**waiting for any device**), Windows needs the driver for this mode too. Repeat the Device Manager steps of section 5 for the new device, again choosing **Android Bootloader Interface**.

- 5 **Check that no OxygenOS update is pending:** type `fastboot getvar snapshot-update-status`. If it answers `snapshot` or `merging`, **stop:** type `fastboot reboot`, let Android start, wait ten minutes, restart Android once more, and begin this section again. Any other answer (`none`, or an error saying the variable is unknown) is fine.
- 6 **Back to the bootloader:** type `fastboot reboot bootloader`.

YOU ARE READY IF

the phone is unlocked, you have written down Android's slot, fastbootd answered yes, no update is pending, and the phone shows the bootloader screen again.

8 Install Antumbra

Keep the cable connected and do not touch the phone while a command runs. Use the commands for **your** slot X from section 7. (On Windows, start each `fastboot` with `.\fastboot.exe`; on macOS and Linux with `./fastboot`.)

8.1 Write the three start-up files (in the bootloader)

Android was on slot a → X = b

```
fastboot flash vbmeta_b antumbra-0.1.0-alpha.2-oneplus-hotdog-vbmeta-disabled.img
fastboot flash dtbo_b antumbra-0.1.0-alpha.2-oneplus-hotdog-dtbo.img
fastboot flash boot_b antumbra-0.1.0-alpha.2-oneplus-hotdog-boot.img
```

Android was on slot b → X = a

```
fastboot flash vbmeta_a antumbra-0.1.0-alpha.2-oneplus-hotdog-vbmeta-disabled.img
fastboot flash dtbo_a antumbra-0.1.0-alpha.2-oneplus-hotdog-dtbo.img
fastboot flash boot_a antumbra-0.1.0-alpha.2-oneplus-hotdog-boot.img
```

Each command ends with OKAY and Finished. If one says FAILED, stop and see section 10.

8.2 Write the system (in fastbootd)

```
fastboot reboot fastboot
fastboot getvar is-userspace
```

The second command must answer `is-userspace: yes`. Then:

```
fastboot -S 128M flash userdata antumbra-0.1.0-alpha.2-oneplus-hotdog-userdata.simg
```

This command sends the 1.5 GB file in pieces of 128 MB. It takes several minutes and prints a line for each piece. **Wait until it prints Finished.** Do not write this file from the bootloader (without `reboot fastboot`): on this phone that stalls.

8.3 Switch to Antumbra's slot and start it

X = b

```
fastboot reboot bootloader
fastboot set_active b
fastboot reboot
```

X = a

```
fastboot reboot bootloader
fastboot set_active a
fastboot reboot
```

THERE IS NO BACKUP STEP, ON PURPOSE

Other guides back up the start-up files first with `fastboot fetch`. This phone does not support that command, and you do not need it: Android's own slot was not touched, which is your way back (section 11).

9 First start

- 1 The phone first shows the **unlocked-bootloader warning**; it continues by itself after a few seconds.
- 2 The screen may stay dark for a while: **the first start can take a few minutes**. Leave the phone alone for up to ten minutes before deciding something is wrong.
- 3 **Antumbra's Welcome screen** appears. Choose your language and keyboard. Optionally create the encrypted **Persistent Storage** (only for files you choose to keep; everything else is forgotten at every shutdown) and switch on **Android apps**. Then press **Start**.
- 4 Connect to Wi-Fi from the top bar. Antumbra connects to **Tor** first; all your connections go through it. Tor Browser is in the app grid.

If there is no Wi-Fi

Wi-Fi needs the modem firmware that Antumbra reads from the phone at every start. To see what it found, open the **Console** app and type:

```
cat /run/antumbra/phone-firmware/status.json
```

It should show "complete": true, with modem and bluetooth found in Antumbra's slot and vendor found (usually in Android's slot, which is normal). If modem is missing, the phone's firmware partitions are empty or were replaced (for example by another custom system); Wi-Fi cannot work then. If only vendor is missing, everything works but graphics are slower.

EVERY START AFTER THIS ONE

Antumbra starts by itself whenever you switch the phone on. Nothing you did in a session remains after a shutdown, except what you saved in Persistent Storage.

Updating Antumbra later

An update writes a new release over Antumbra's slot X and erases Persistent Storage, so copy out what you need first. Repeat sections 6 to 8 with the new files and the **same X**, the letter that is not Android's on your note. Do not work out X from `current-slot` again: it now shows Antumbra's slot, and "the other one" would be Android's. (The Linux script remembers Android's slot for each phone: `build/flash.sh --release FOLDER`.)

10 If something goes wrong

Two key combinations solve most problems. **Switch the phone off:** hold **Power** + **Volume Up** for about ten seconds. **Start the bootloader:** from off, hold **Power** + **Volume Up** + **Volume Down**.

What you see	What to do
<code>fastboot devices</code> prints nothing, or a command waits forever	Try another cable and another USB port (a USB 2.0 port works best). On Windows install the driver (sections 5 and 7). On Linux try <code>sudo</code> . Make sure the phone shows the bootloader or fastbootd screen.
FAILED when writing a start-up file	Check that you typed the slot letter and file name exactly, and that <code>fastboot getvar unlocked</code> says yes. Nothing harmful happened to Android's slot; you can run the command again.
The large file stops with an error	Make sure you are in fastbootd (<code>fastboot getvar is-userspace</code> says yes) and that no update is pending (section 7, step 5). Then run the <code>-S 128M flash userdata</code> command again.
After installing, the phone returns to the bootloader, shows a red error screen, or stays dark for more than ten minutes	Switch it off, start the bootloader, and either try again with <code>fastboot set_active X</code> and <code>fastboot reboot</code> (this resets the phone's count of failed starts), install again into the same X (sections 7 and 8, using the letter on your note, not <code>current-slot</code>), or go back to Android (section 11). Do not wait for the phone to switch back to Android by itself: on this model it may not.
Android starts and asks to erase data, or the OnePlus recovery appears	The phone started Android's slot. Either install again from section 7, or let Android erase its data and keep using Android (section 11).
No Wi-Fi, no sound or slow graphics in Antumbra	See "If there is no Wi-Fi" in section 9.
The computer shows QDLoader: 9008, QUSB_BULK or 05c6:9008	The phone is in its emergency download mode. Do not run any command. Disconnect it, switch it off with the key combination, and start the bootloader. If that never works again, see section 12.

11 Going back to Android

Android's slot was never written, so you can return to it. Its data, and everything in Antumbra, is erased in the process.

- 1 Switch the phone off and start the bootloader (**Power** + **Volume Up** + **Volume Down**).
- 2 Type `fastboot set_active` followed by **Android's** slot, the letter on your note from section 7 (for example `fastboot set_active a`).
- 3 With the **Volume** keys choose **Recovery mode** on the bootloader screen and press **Power**. In the recovery, choose the option that wipes or formats data (a factory reset) and confirm.
- 4 Restart. Android starts as it was before you installed Antumbra, without your data. Antumbra's files stay in the other slot, where they do no harm.

DO NOT SKIP THE FACTORY RESET

Without it, Android finds Antumbra's data in place of its own and erases it anyway, and the first start can take very long.

12 Last resort: the MSM Download Tool

If the phone no longer reaches the bootloader at all, or neither slot starts, the **OnePlus MSM Download Tool** rewrites the entire phone with OxygenOS from its emergency download mode (EDL). It is OnePlus's own service tool; OnePlus has never published it officially, and copies circulate in the community (for this phone mainly through the XDA Developers forum thread "OP7TPRO Unbrick tool"). Be careful where you download it from.

- It runs on **Windows only** and needs the Qualcomm USB driver; the phone must show up as Qualcomm HS-USB QDLoader 9008.
- You need the package for your exact model and region (HD1913 is the "EU" variant). The packages contain OxygenOS 10 or 11; update to F.22 afterwards.
- To enter EDL: switch the phone off, hold **Volume Up** + **Volume Down** and connect the cable.
- It erases everything and locks the bootloader again. Never use its "SMT" mode: it can erase the phone's IMEI.

A Quick reference

Keys (from a switched-off phone)

Switch off (when on)	Power + Vol Up , 10 s
Bootloader	Power + Vol Up + Vol Down
Recovery	Power + Vol Down
EDL (MSM tool only)	Vol Up + Vol Down , then connect USB

What the checks must answer

unlocked	yes
current-slot	Android's slot, before the first install only
is-userspace	yes, in fastbootd
snapshot-update-status	not snapshotted or merging

The whole install (X = Antumbra's slot)

```
fastboot getvar unlocked
fastboot getvar current-slot
fastboot reboot fastboot
fastboot getvar is-userspace
fastboot getvar snapshot-update-status
fastboot reboot bootloader
fastboot flash vbmeta_X antumbra-0.1.0-alpha.2-oneplus-hotdog-vbmeta-disabled.img
fastboot flash dtbo_X antumbra-0.1.0-alpha.2-oneplus-hotdog-dtbo.img
fastboot flash boot_X antumbra-0.1.0-alpha.2-oneplus-hotdog-boot.img
fastboot reboot fastboot
fastboot getvar is-userspace
fastboot -S 128M flash userdata antumbra-0.1.0-alpha.2-oneplus-hotdog-userdata.img
fastboot reboot bootloader
fastboot set_active X
fastboot reboot
```

X is Antumbra's slot: the letter that is not Android's, from your note (on the very first install, the one `current-slot` did not name).
On Windows type `.\fastboot.exe`, on macOS and Linux `./fastboot`.

ABOUT THIS GUIDE

Antumbra is free software (GPL-3.0-or-later), not affiliated with or endorsed by OnePlus, Google, the Tor Project or the Tails project. Its images contain no OnePlus or Qualcomm software; the phone's firmware is read from the phone itself. The steps follow the hardware-tested procedure of the hotdog-linux-bringup port, with changes for Antumbra described in its `docs/flashing.md`. Facts about this phone that could not be confirmed (such as the exact wording of its screens) are described in general terms. Nothing here is legal advice, and you use Antumbra at your own risk.